The cost of State Pension guarantees under forecast uncertainty*

Vahid Ahmadi† Max Ghenis‡

November 2026

Abstract

Rules that uprate by the highest of several indices each year have a cost that depends on the indices' joint path. We study the adjusted Triple Lock, announced in the UK's 2026 Budget: from April 2030 the State Pension rises by the highest of 2.5%, CPI and the amount needed to keep its April 2029 value relative to earnings. Its gap from the triple lock is driven by a reflected random walk and grows only when earnings lead after prices or the floor led. Started in April 2012, it would have left the pension 6.2% lower by 2026-27. In 2039-40 it lowers State Pension spending by £0.5bn on the Office for Budget Responsibility's (OBR) central path; by £10.9bn under the OBR's long-term triple-lock convention, an upper bound; and by £11.4bn or £11.9bn averaged over 160 simulated paths with earnings growth 0.5 points lower or higher. The government estimates £15bn for Great Britain. The model of prices and earnings failed one condition of a test on past forecasts that we registered before scoring, so these are scenarios, not an expected value. On the OBR long-term path, losing households have £801 a year less, and pensioner poverty after housing costs rises by 0.70 to 0.75 points, and up to 1.52 points on a simulated path with a large gap.

Keywords: pension indexation; triple lock; path dependence; microsimulation; forecast evaluation; proper scoring rules; registered analysis plans
JEL codes: H55, H68, C53, D31

*Acknowledgements: We thank María Juaristi and colleagues at PolicyEngine for comments.
†PolicyEngine. Email: vahid@policyengine.org.
‡PolicyEngine. Email: max@policyengine.org.

1 Introduction

Indexation rules set how pensions in payment keep pace with prices and earnings, and with them much of a pension system's long-run cost [Vidal-Meliá et al., 2009, OECD, 2023]. A rule that pays the highest of several indices each year is asymmetric: it passes on every year in which one index leads and keeps that gain in all later years. This paper asks how much of the fiscal effect of removing such a ratchet comes from year-to-year variation in the indices, and what can be reported when the model of that variation fails a test against past forecasts.

Since April 2011 the triple lock has raised the State Pension each April by the highest of prices, earnings growth and 2.5% (prices measured by RPI in its first year and by CPI since). Because it takes the highest of three figures every year, a year in which prices or the floor run ahead of earnings raises the pension for good, and later rises build on the higher level. The OBR's long-term projections assume that the triple lock raises the pension 0.6 percentage points a year faster than earnings [Office for Budget Responsibility, 2026c], and it estimates that by 2029-30 the triple lock will have added £15.5 billion a year to spending relative to earnings uprating since its introduction [Office for Budget Responsibility, 2025].

The government announced on 29 September 2026 that it would keep the triple lock for each rise through April 2029 and adjust it from April 2030, and DWP published an analysis of the change the same day [Department for Work and Pensions, 2026b]. The Budget of 28 October 2026 announced the change as the adjusted Triple Lock, to take effect from the next Parliament: from April 2030 the State Pension rises "by the highest of 2.5%, inflation, or the amount required to maintain its value relative to earnings as reached in April 2029" [HM Treasury, 2026, para. 5.23 and Box 5.A]. The change requires amendment of section 150A of the Social Security Administration Act 1992 (Section 3.1). The Budget adds that "It removes the 'ratchet' effect, whereby the State Pension rises ever higher relative to earnings" [HM Treasury, 2026, chapter 1], and puts the saving at: "Adjusting the Triple Lock is estimated to reduce State Pension spending by £15 billion a year by the late 2030s, and by £50 billion a year by the late 2040s" [HM Treasury, 2026, para. 1.29]. Both official documents recognise that this saving depends on the path of the indices. The Budget states: "Under the Triple Lock, the cost of the State Pension depends on which of prices, earnings or 2.5% is highest in each year, and on how those differences compound over time. That makes it inherently hard to forecast" [HM Treasury, 2026, Box 5.A]. The OBR writes that its estimates "depend critically on the order, level and covariation of wage growth and inflation over time" [Office for Budget Responsibility, 2026a, para. 3.53]. We quantify that dependence.

The adjusted Triple Lock keeps the triple lock's protection against prices and the floor but removes the ratchet relative to earnings, so it isolates the part of the triple lock's cost that comes from the variability of the indices. We estimate its effect against the triple lock from 2027-28 to 2039-40 with full runs of PolicyEngine UK [Ahmadi et al., 2026] on the whole survey population, on the OBR's central path, under the OBR's long-term triple-lock convention and on simulated paths of prices and earnings.

The paper makes three contributions. First, we show that the earnings-anchored rule's excess over its earnings path is a reflected random walk, a Lindley recursion [Lindley, 1952], so its gap from the max-of-indices rule depends on the running minimum of the cumulative difference between the indices, as a lookback option's payoff does [Goldman et al., 1979]. The identity is standard; applied to indexation it gives the gap's yearly increment, an upper bound by the smaller of the two indices' cumulative leads, and the exact condition for a positive saving: a year in which prices or the floor lead, followed later by a year in which earnings lead

(Section 3.3). Second, we measure how much of the gross saving the means-tested offsets and income tax absorb, and where the remainder falls: on the OBR long-term path the government keeps 68% of the gross saving, and the loss relative to income is largest in the lower-middle deciles (Section 5.4). Third, we test the model of prices and earnings on past OBR forecasts with proper scoring rules [Gneiting and Raftery, 2007, Scheuerer and Hamill, 2015], under a pass rule and a reporting rule registered before scoring [Casey et al., 2012, Olken, 2015], and show what a failed condition leaves: a scenario envelope rather than an expected value.

The saving is small on the OBR central path, on which earnings lead in almost every year, and much larger on simulated paths whose inputs vary from year to year, although those paths also move average earnings growth away from the OBR's (Section 5.1).

Section 2 describes the triple lock's record and relates the paper to the literature. Section 3 states the two rules, derives where the saving comes from and replays the rule on past inputs. Section 4 describes the data, the microsimulation, the paths of prices and earnings and the registered test. Section 5 gives the fiscal effects, the mechanism along single paths and the distributional effects. Section 6 compares the results with official costings, Section 7 discusses what they imply for costing such rules, and Section 8 concludes and states the main limitations.

2 Background and related literature

2.1 The triple lock since 2011

Figure 1 shows the rises actually paid in each April from 2011 to 2026. April 2011's rise, 4.6%, was set by RPI under that year's up-rating order [Social Security Benefits Up-rating Order, 2011], while the replay of the CPI rule gives 3.1%. From 2012, CPI set the rise five times, earnings six times and the floor four times. In April 2022 the law suspended the earnings leg, whose input was 8.4% on today's figures, so that year's rise was CPI, 3.1% [Social Security (Up-rating of Benefits) Act, 2021]. Over the 16 rises the pension rose by 89.4% as paid.

The rest of this subsection replays the rule on today's revised ONS figures, so that both rules see the same inputs. The replay differs from the rise paid in April 2016, 2019, 2020 and 2025, by 0.1 to 0.3 point, because earnings figures have since been revised, and in April 2011. On the replay the pension rose 87.4%, against 66.2% for earnings alone and 60.2% for prices alone, which leaves it 12.7% above where an earnings link alone would have taken it.

2.2 Related literature

Automatic adjustment mechanisms link a pay-as-you-go pension system's benefits or contributions to indicators, so that its finances respond to demographic and economic change without new legislation [Vidal-Meliá et al., 2009]; Knell [2010] shows how the choice of adjustment factor changes the return such a system pays. The indexation of pensions in payment is one such link, and the OECD's review of the inflation surge since 2021 records how far different rules protected pensioners' purchasing power [OECD, 2023]. The triple lock is unusual in taking the highest of three figures each year, and Portes [2023] places it in the longer history of the State Pension's value relative to earnings. A rule that pays the higher of two growth rates and compounds the result has the structure of a path-dependent option: each year adds the positive part of the difference between the indices, and the adjusted rule's return to its earnings path makes the gap between the rules depend on the running minimum of the cumulative difference, as the payoff of a lookback option depends on the running minimum of a price [Goldman et al., 1979].

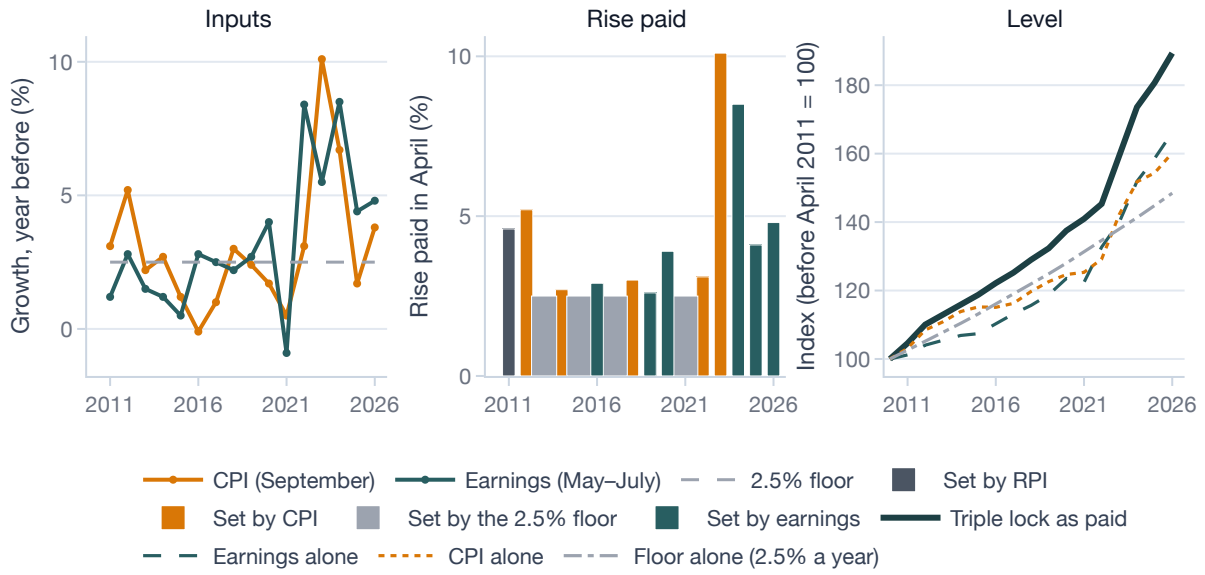


Figure 1: The triple lock since 2011: the rises paid, and a level that ended well above earnings alone. Left: September CPI and May–July earnings growth in the year before each April (today's ONS figures), and the floor. Centre: the rise paid, coloured as the input that set it. Right: the level as paid, against earnings alone, CPI alone and the floor alone (2.5% a year).

The option analogy connects the paper to the valuation of guarantees. Merton [1977] values a loan guarantee as a put option, Lucas [2012] argues that public guarantees and policies should be valued at market prices that charge for the risk they carry, and Geanakoplos and Zeldes [2010] apply that approach to accrued Social Security benefits. In pension finance, de Jong [2008] values the liabilities of a fund that indexes benefits only when its funding allows, an option embedded in the indexation rule. We compute the expected cost in cash, undiscounted and with no charge for risk, because that is the measure in which official costings of uprating rules are stated and the paper compares with them; a market value of the triple lock's guarantee would also need a price for UK earnings and inflation risk, which we leave aside. We use the option analogy for the structure of the payoff only.

Long-run projections of pension spending are uncertain, and stochastic projections of social security finances express that uncertainty as distributions over paths of demographic and economic inputs [Lee and Tuljapurkar, 1998]. The OBR's fiscal risks and sustainability reports treat the triple lock as a source of fiscal risk and project its cost against earnings uprating [Office for Budget Responsibility, 2025]; in its 2026 long-term scenarios the triple lock accounted for around a third of the projected rise in State Pension spending as a share of GDP [Office for Budget Responsibility, 2026a, para. 3.48]. Central banks and fiscal analysts publish fan charts to show the uncertainty around a central projection [Britton et al., 1998, Celasun et al., 2006]. A model that supplies such a distribution can be tested against outcomes: probability integral transforms check its calibration [Diebold et al., 1998], a forecast should be as sharp as its calibration allows [Gneiting et al., 2007, Mitchell and Wallis, 2011], and proper scoring rules reward a forecaster for reporting its true predictive distribution [Gneiting and Raftery, 2007],

including for multivariate outcomes [Scheuerer and Hamill, 2015]. Density forecasts of inflation and output are better calibrated when they allow volatility to change over time [Clark, 2011], which our model does not.

Static tax-benefit microsimulation applies the rules of the tax and benefit system to a representative survey of households, with behaviour held fixed, and is the standard tool for the distributional analysis of benefit reforms [Sutherland and Figari, 2013]; UKMOD is one such model for the four nations of the UK [Richiardi et al., 2021]. Dynamic microsimulation models instead project individual life courses, including pension entitlements, forward in time [Li and O'Donoghue, 2013]; DWP's costing of the adjusted Triple Lock uses one (Section 6). The net saving depends on how many newly entitled pensioners claim Pension Credit, and take-up of means-tested benefits responds to the size of the entitlement and the cost of claiming [Pudney et al., 2006]; our runs hold each benefit unit's claim decision fixed. Pre-analysis plans fix a study's hypotheses and tests before the data are seen, so that results cannot be chosen after the fact [Casey et al., 2012, Olken, 2015], and Christensen and Miguel [2018] review their spread in economics. They are rare in model-based policy costing, where the analyst chooses both the model and its evaluation; Section 4.4 describes the plan we registered and how it changed.

3 Policy rules

3.1 The two rules

Under the triple lock the full basic and new State Pension rise each April by the highest of three figures: the CPI 12-month rate for the previous September [Office for National Statistics, 2026b]; growth in average weekly earnings, whole-economy total pay, in May to July of the previous year [Office for National Statistics, 2026d]; and 2.5%. ONS publishes both inputs to 0.1 point, and both rules here use them at that precision. We index each input by the April it sets: c_t is the September CPI rate and e_t the May–July earnings growth published before April t, and f is the 2.5% floor. Fiscal year t pays the level set in April t.

The adjusted Triple Lock follows the triple lock to April 2029. From April 2030 let L_t be the pension's level and A_t an earnings path that starts at the 2029-30 level and grows with May–July earnings, $A_t = A_{t-1}(1 + e_t)$. Then

$$L_t = \max\big(L_{t-1}\,(1 + \max(c_t, f)),\ A_t\big).$$

The pension never rises by less than the higher of CPI and the floor, and never falls below its earnings path. In April 2030 this rise equals the triple lock's, since the earnings path starts at the same level, so the two rules can first differ in April 2031 and the saving in 2030-31 is zero on every path. As a modelling assumption we round any top-up to the earnings path up to the next 0.1 point, so the pension never ends below that path; this favours the pensioner by at most 0.1 point a year.

The Budget's wording fixes the anchor: the pension's value relative to earnings "as reached in April 2029". DWP's description says the same in other words: the pension will "reach a record high relative to earnings over the coming three years", and from April 2030 it keeps pensioners benefiting "via the State Pension retaining that record high value relative to earnings": "In any given year it will go up by at least inflation or 2.5% – and anything more that is needed to retain that value". It adds: "This means that over time the State Pension is set to rise in line with average earnings" [Department for Work and Pensions, 2026b]. Two choices remain ours. The

rule uses the triple lock's measures, September CPI and May–July earnings, and the top-up is rounded up, as above.

The law requires only that the basic and new State Pension rise at least in line with earnings [Social Security Administration Act, 1992]; the prices leg and the floor are policy on top. In a year when the adjusted Triple Lock rises by the higher of CPI and the floor while its level returns towards its earnings path, its rise is below earnings growth, which section 150A does not allow, so the rule needs primary legislation, as the suspension of the earnings leg for April 2022 did [Social Security (Up-rating of Benefits) Act, 2021]. On both OBR scenarios this happens once, in April 2032; on the model paths near the median and the 90th percentile of the gap, in four and five of the Aprils from 2031 to 2039.

3.2 Where the saving comes from

When earnings lead, both rules give the same rise. The rules part only after a year in which the higher of CPI and 2.5% beats earnings. Both pay that higher figure, but it lifts the adjusted Triple Lock above its earnings path. When earnings lead again, the triple lock rises with earnings from the higher level, while the adjusted Triple Lock rises only by the higher of CPI and the floor until its earnings path catches up. Under the triple lock each such rise stays in the pension's level; under the adjusted rule the level returns to its earnings path over the following years.

A two-year example shows the arithmetic. Suppose CPI is 6% and earnings growth 2% in the first year, then CPI 2% and earnings 4%. The triple lock rises by 6% and then 4%, 10.24% in all. The adjusted Triple Lock also rises by 6% in the first year, which leaves it 3.85 log points above its earnings path, and then by the floor, 2.5%, 8.65% in all. In the second year earnings lead by 1.45 log points, less than the excess, so the whole lead becomes the gap: the adjusted rule ends 1.4% below the triple lock (1.59 points less cumulative growth). The next subsection states this in general.

3.3 A formal statement

Write $m_t = \max(c_t, f)$ for the higher of CPI and the floor that applies in April t, and T_0 for April 2029, the last rise under the triple lock alone. With P_t the triple lock's level and L_t and A_t as above,

$$P_t = P_{t-1}\big(1 + \max(m_t, e_t)\big), \qquad L_t = \max\big(L_{t-1}\,(1 + m_t),\ A_t\big), \qquad A_t = A_{t-1}\,(1 + e_t),$$

with $P_{T_0} = L_{T_0} = A_{T_0}$. Let $d_t = \ln(1 + m_t) - \ln(1 + e_t)$, positive in a year in which CPI or the floor beats earnings, and measure each rule's level against the earnings path, $y_t = \ln(P_t/A_t)$ and $x_t = \ln(L_t/A_t)$. Then

$$y_t = y_{t-1} + \max(d_t, 0), \qquad x_t = \max(x_{t-1} + d_t,\ 0), \qquad y_{T_0} = x_{T_0} = 0.$$

The triple lock accumulates every positive d_t, while the adjusted rule's x_t is a cumulative sum reflected at zero, a Lindley recursion [Lindley, 1952]. With $D_t = \sum_{s=T_0+1}^{t} d_s$ and $D_{T_0} = 0$, the log gap between the rules, $G_t = y_t - x_t$, is

$$G_t = \sum_{s=T_0+1}^{t} \max(-d_s, 0) + \min_{T_0 \le s \le t} D_s, \qquad \Delta G_t = G_t - G_{t-1} = \min\big(\max(-d_t, 0),\ x_{t-1}\big) \ge 0,$$

and

$$0 \le G_t \le \min\Big(\sum_{s=T_0+1}^{t} \max(d_s, 0), \sum_{s=T_0+1}^{t} \max(-d_s, 0) \Big).$$

Proposition. The gap never falls, and it grows only in a year in which earnings lead while the adjusted rule is still above its earnings path, by the smaller of earnings' lead and the remaining excess. The gap after April t is positive if and only if some year $s \le t$ in which prices or the floor lead ($d_s > 0$) is followed by a later year $r \le t$ in which earnings lead ($d_r < 0$). In particular the gap is zero on any path on which earnings lead in every year and on any path on which they never lead, and a lead for prices or the floor in the final year alone gives no saving. The first appendix gives the proof.

Because $G_t \ge 0$ on every path, a path on which earnings lead in every year after the switch gives the smallest possible saving, zero. A path close to that, such as the OBR's central path, gives a saving close to zero, and a distribution of paths with changes of lead gives a positive average. The proposition does not say that variation around a given path raises the saving above that path's own: G_t adds a convex term in the yearly differences to a concave one, so a spread around a path with a positive gap can lower it. The triple lock's log level is convex in each year's input difference, so variability raises it on average; the adjusted rule's gap has no such property, and the size of the saving on varied paths is an empirical question that Section 5 answers. The increment also says when the saving appears: larger earnings leads use up the excess x_{t-1} faster, so higher earnings growth after a year of high inflation brings the saving forward.

The gross saving in a fiscal year is the difference in basic and new State Pension paid,

$$S_t^g = \sum_i w_{it} \left(b_{it}^{\mathrm{TL}} - b_{it}^{\mathrm{A}} \right) \approx \left(1 - e^{-G_t} \right) B_t^{\mathrm{TL}},$$

where w_{it} is the survey weight of person i, b_{it} their basic or new State Pension under each rule and B_t^{TL} total spending on both under the triple lock; the approximation is exact, up to rounding, when each person's amount is a fixed share of the full rate. The net saving is the change in the government's direct budget balance, $S_t^n = \sum_h w_{ht} \left(\tau_{ht}^{\mathrm{A}} - \tau_{ht}^{\mathrm{TL}} \right)$, where w_{ht} is the weight of household h and τ_{ht} the direct taxes it pays less the benefits it receives, State Pension included. The rounding of the top-up to the earnings path adds a small non-negative term to x_t, so the identities hold up to that rounding in the model runs.

3.4 The rule on past inputs

The decomposition can be checked on the inputs that have already occurred, with no model of their future path. Table 1 and Figure 2 replay the adjusted Triple Lock on today's figures from each earlier April. Had it started in April 2012, 2013, 2014 or 2015, its level in 2026-27 would be 6.2% below the triple lock's on the same inputs. Applied to the full new State Pension actually paid, £12,548 a year, that would be £11,769. Most of the difference follows April 2023, when September CPI of 10.1% set a rise far above earnings growth of 5.5%; the gap opened in the earnings-led Aprils after it, as the increment formula says. Started in April 2024, 2025 or 2026, the rule would have paid the same as the triple lock every year, because earnings have led in every April since. In full model runs, State Pension spending in 2026-27 would have been £7.8bn lower, and the government £5.6bn better off once tax and benefits respond. Averaged over the Aprils from 2012 to 2026, the pension would have been 1.6% below the triple lock. The OBR replays the rule on outturn from 1990 to 2025, anchored in 1990: "Averaging over the

period as a whole, state pension payments would have been on average 1.4 per cent lower each year than under the current triple lock and 4.4 per cent higher than under earnings uprating" [Office for Budget Responsibility, 2026a, para. 3.51].

Table 1: The adjusted Triple Lock replayed on today's figures from earlier starts: how far its level ends below the triple lock's, the changes of lead between earnings and the higher of CPI and the floor in the Aprils from the start to the last, and the Aprils in which it rose by less than the triple lock (rule arithmetic).

Start	Level below the triple lock, 2026-27	Changes of lead	Aprils with a smaller rise
April 2012, 2013, 2014 or 2015	6.2%	five	2016, 2019, 2020, 2024, 2025 and 2026
April 2016, 2017 or 2018	5.1%	four	2019, 2020, 2024, 2025 and 2026
April 2019, 2020 or 2021	4.4%	two	2024, 2025 and 2026
April 2022 or 2023	4.2%	one	2024, 2025 and 2026
April 2024, 2025 or 2026	0.0%	0	none

Note: Starts in the same row give the same levels; changes of lead are counted from the earliest.

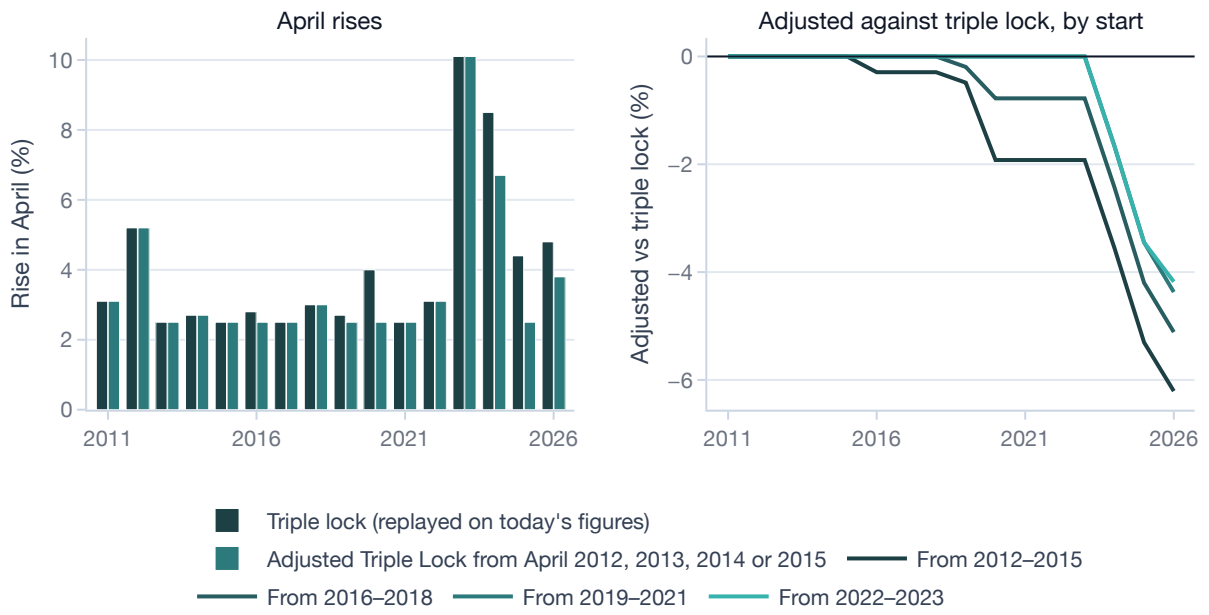


Figure 2: Had the adjusted Triple Lock started earlier, the pension would now be lower, mostly because of one inflation spike. Left: the April rises under the triple lock and under the adjusted rule from the earliest start that changes anything, both replayed on today's figures. Right: the adjusted rule's level against the triple lock's, by start year.

4 Data and method

4.1 Model and data

Every saving and household figure for a given path is a full run of PolicyEngine UK [Ahmadi et al., 2026, PolicyEngine, 2026b,c] on the whole survey population, once under each rule. PolicyEngine UK is a static tax-benefit microsimulation model of the kind Sutherland and Figari [2013] describe: it applies the tax and benefit rules of each year to survey households and allows no behavioural response. All runs use policyengine 6.11.1, which pins policyengine-uk 2.151.0 and policyengine-core 3.33.3, with the national Microcosm dataset for 2024-25 published with it (78,234 households) [PolicyEngine, 2026a]. The Enhanced Family Resources Survey, version 1.58.2, runs only for the check against DWP's caseloads (Table 8). A path's CPI and earnings growth replace the model's economic assumptions before the data load, so benefit rates, tax thresholds and earnings follow it, and every run checks that they do. Only the basic and new State Pension differ between the two runs of a path. Under both rules the additional State Pension rises with September CPI and State Pension age follows the law's timetable. Pension-age Housing Benefit is modelled as under current law. Money is in cash terms for the UK unless stated.

Three conventions shape the net saving. First, the Pension Credit guarantee rises with May–July earnings, the minimum section 150A requires, under both rules. This is the statutory minimum; in practice the guarantee has also risen by the same percentage as the State Pension, as in April 2023, when both rose by 10.1% against earnings growth of 5.5% [Social Security Benefits Up-rating Order, 2023]. If the guarantee instead rose by the same percentage as the State Pension under each rule, it would be lower under the adjusted rule, the Pension Credit offset would be smaller and the net saving larger. Second, the income tax personal allowance stays at £12,570 to 2030-31, as current law sets it, and then rises with each path's September CPI; from April 2031, when the rules can first differ, the full new State Pension is above it, so a lower pension means less income tax even for a pensioner with no other income. The Budget's Pension Tax Allowance removes income tax for pensioners whose total income is at or below the full new State Pension from April 2027 to April 2030 [HM Treasury, 2026, Table 6.1]; it lapses before the two rules can first differ, so it does not affect the differences reported here, and we assume no successor. Third, Pension Credit take-up follows the dataset's own claim decision for each benefit unit, so a household newly entitled under the adjusted rule claims only if that decision says it would; Housing Benefit and council tax reduction respond only for households already receiving them, and passporting from the guarantee credit follows actual receipt.

4.2 The population in each year

The dataset represents households in 2024-25, and each record stands for a person of its age in every year. Through 2025, the year the dataset calibrates to, each record keeps its own weight; after it, household weights are reweighted so that each age and sex group grows as in the ONS's 2024-based population projection [Office for National Statistics, 2026e]. Ages are not top-coded, so every recorded age is kept. Each person's State Pension type follows from the date of birth that their age implies in each year: those who reach State Pension age before 6 April 2016 are on the basic State Pension, later cohorts on the new one [Pensions Act, 2014]. A record of a given age therefore moves from the basic to the new State Pension as later cohorts reach that age. Deaths, migration and contribution histories are not simulated, and pensioners abroad are outside the dataset. Records carried forward from today's pensioners may understate the entitlements of people reaching State Pension age in the 2030s, most of whom will have spent

their whole working lives under the new State Pension.

4.3 Paths of prices and earnings

The OBR central path uses the OBR's October 2026 Economic and fiscal outlook, published with the Budget, for calendar years to 2030 [Office for Budget Responsibility, 2026a], and the OBR's March 2026 long-term economic determinants after that [Office for Budget Responsibility, 2026c], which the OBR describes as a projection, not a forecast; the OBR published no long-term determinants with its October forecast. We call the combination the OBR central path. April 2027's State Pension rise uses the published May–July 2026 earnings growth, 3.9%, and September 2026 CPI, 3.5% [Office for National Statistics, 2026d,b]. For the Aprils 2028 to 2031 the inputs are the OBR's September-quarter CPI and April–June earnings growth; later Aprils use its calendar-year figures. The OBR forecasts average earnings on a national-accounts measure, which stands in here for May–July average weekly earnings. Both rules share these inputs.

The simulated paths come from a model of monthly changes in the CPI index [Office for National Statistics, 2026a] and in average weekly earnings [Office for National Statistics, 2026c], fitted to data from 2000 to the latest published month, with the furlough months of 2020–21 left out. Each month's change depends on the previous month's changes in both series, with fixed coefficients; the shocks are past pairs of residuals drawn at random, so prices and earnings keep their joint variation, but the model has no periods of higher or lower volatility and no uncertainty about its coefficients. From the simulated months come both the statutory inputs (September CPI, May–July earnings growth) and the calendar-year growth the rest of the model uses. Every path is then shifted by the same smooth drift so that the average calendar-year growth matches the OBR's forecast in every year, while each path keeps its own ups and downs; the shift does not force the statutory measures to match the central path's statutory inputs, and we do not report how far their averages differ. The earnings scenarios move that drift by 0.5 points a year from 2031.

On the simulated paths the lead between September CPI and May–July earnings changes 0.43 times a year. In the inputs that set the April rises of 2011 to 2026 it changed 0.33 times a year with April 2022 as in law and 0.47 times on the published earnings figure for that year; on calendar-year CPI and average earnings from 1989 to 2025 it changed 0.17 times a year. The OBR central path has one change of lead between earnings and the higher of CPI and the floor from April 2030 to 2039. The simulated rate is thus above the record since 2011 and well above the longer record, which is dominated by years in which earnings led; we have not tested how the rate depends on the window the model is fitted to.

The fiscal averages for the earnings scenarios do not average every draw. The model makes 50,000 equally weighted draws; we sort them into ten groups by the size of the gap between the rules in 2039-40, plus a group in which the rules pay the same, and run 160 of them in full, allocated across the groups in proportion to each group's size and spread. Each average is then a stratified estimate of the mean over all 50,000 draws, and its margin is a 95% Monte Carlo interval from a standard error that combines the sampling of the full runs with the sampling of the draws. The margin is conditional on the model and leaves out uncertainty about its coefficients. The two earnings scenarios use the same draws, shocks and full-run selection, shifted in opposite directions.

4.4 Testing the model on past forecasts

An average over the simulated paths needs a model whose paths behave like the real inputs, so we registered a test on past OBR forecasts [Office for Budget Responsibility, 2026b] before scoring any model. Five versions of the model of prices and earnings are fitted only on data available before each forecast, centred on that forecast, and scored, with proper scoring rules for each statistic and for the joint outcome [Gneiting and Raftery, 2007, Scheuerer and Hamill, 2015] and with the coverage of central bands, on the gap between earnings and CPI in each year, on how often the lead changes and the floor binds, and on the gap the two rules open after four rises. One test uses seven forecasts, one a year from the March 2016 EFO to the March 2022 EFO; another uses thirteen, from the June 2010 Budget forecast to the March 2022 EFO. Table 2 gives the pass rule; the thresholds were fixed together before any score was seen.

The rule changed once, after the first scores and before the second. Under the first rule every version failed, because April 2022 was scored on the published earnings: each had only one of six four-year outcomes inside its band, because the furlough schemes distorted 2021 pay growth. The second rule differs from the first in two elements, both fixed after we had seen the first scores. It scores April 2022 as the law applied it, with the earnings leg suspended. And it changes what follows a failure of the main version. The first rule would then have taken the passing version with the lowest average ratio of its scores to the main version's; on the second rule's scores that is Monthly, two lags, which passes, so the first rule's fallback would have authorised an average over its paths. The second rule instead forbids promoting another version on its scores: only a passing main version may supply an average, and if the main version fails the analysis reports scenarios only. Whatever the verdict, the second rule keeps four scenarios: the OBR central path, the OBR long-term path, the historical replay and the paired paths with earnings growth 0.5 points lower and higher. We registered the second rule before scoring under it; the rest of the first rule is unchanged.

On the Budget's inputs the main version failed one condition (Table 9, Table 10). It expected earnings growth below 2.5% in 44.6% of the years tested on the seven-forecast test, against 17.9% in practice, about 2.5 times as often, on the event that sets $d_t > 0$. That average error, 0.267, exceeds the limit of 0.25 by 0.017, which is within one standard error; but the error itself is 2.6 standard errors from zero, using a standard error of 0.101 that allows for the overlap between four-year windows. All five versions share the error, with floor biases from 0.243 to 0.275, which points to a cause they have in common, such as centring every path on the OBR's point forecasts or the shared pool of residuals; the test does not separate the two. The main version passed every other condition. The outcome fell inside its central 80% band in five of seven four-year windows and 83% of annual checks. It expected 1.06 changes of lead per four-year window against 1.00 realised; on the published 2022 earnings, a sensitivity that does not enter the verdict, 1.29 against 1.86. The realised gap since 2011 sits at the 55th percentile of its paths. Two other versions passed and two failed on the same condition, but the rule forbids promoting another version, so this paper gives no expected value.

The test has little power. With seven overlapping four-year windows, a model whose bands cover far less than 80% could still pass, and because the upper limit on coverage is 100%, the coverage condition cannot detect bands that are too wide. A third registered rule applies only to estimates that use OBR forecasts published after this paper's inputs, and adds a test on outcomes not yet observed. On simulated outcomes a correct model passes that test 35.8% of the time after four years, 68.5% after eight and 85.6% after twelve, and a model that states double the true rate of earnings growth below the floor passes 35.1%, 43.1% and 63.7%, so four

years cannot tell the two apart.

Table 2: The pass rule for the test on past forecasts, April 2022 scored as the law applied it.

Condition	Threshold
Coverage	Between 50% and 100% of outcomes, yearly and after four years, fall inside the central 80% band
Coverage, plausible range	The range of plausible coverage after four years includes 80%
Bias in the yearly gap	Average error in earnings minus CPI at most 1.5 points
Bias in lead changes	Average error in the number of lead changes at most 1.0 a forecast
Bias in floor years	Average error in the share of years in which the floor binds at most 0.25
Accuracy scores	No score more than 25% worse than the main version's
Past years	The realised gap since 2011 between the 5th and 95th percentiles of its paths

Note: A version passes only if it meets every condition on both tests.

5 Results

5.1 Fiscal effects on four scenarios

Table 3 and Figure 3 give the reduction in State Pension spending under the adjusted Triple Lock, relative to the triple lock, on four scenarios. Two are single full runs. One is the OBR central path. The other, the OBR long-term path, keeps the central path's inputs but pays the triple lock at the OBR's long-term triple-lock rate from April 2032, which phases in to earnings growth plus 0.6 points, and applies the adjusted rule to the same inputs. The other two take 160 simulated paths of prices and earnings, each a full run under both rules, and move calendar-year earnings growth 0.5 points a year lower or higher from 2031 (Section 4.3). We attach no probability to any of them, and the four do not form a probability interval.

On the OBR long-term path the saving is £10.9bn in 2039-40, £7.3bn once income tax and benefits respond: extra Pension Credit offsets £0.8bn and lower income tax £2.4bn. The adjusted rule's level on this path is the same as on the central path, £18,840 a year for the full new State Pension in 2039-40, because the two runs share their inputs; only the triple lock differs. Earnings lead CPI and the floor in every April from 2032, so the adjusted Triple Lock returns to its earnings path and then follows it, and the whole of the OBR's premium counts as saving. The figure is therefore the premium's compounded cost: G_t is the sum of the log differences between the OBR's long-term triple-lock rate and earnings growth from April 2032, 4.34 log points by 2039-40, and $(1 - e^{-G_t})B_t^{\mathrm{TL}}$ gives 4.25% of £255.8bn in basic and new State Pension spending. The premium is the OBR's average allowance for years in which CPI or the floor beat earnings, and no path of CPI and earnings pays it every year. Because the adjusted rule is given no premium of its own on this path, the scenario is an upper bound on the saving under the OBR's convention: the OBR's replay of 1990–2025 leaves the adjusted rule above earnings

uprating on average [Office for Budget Responsibility, 2026a, para. 3.51], and its provisional long-term estimate leaves debt above an earnings-uprating scenario because it expects periods in which inflation or the floor would still bind [Office for Budget Responsibility, 2026a, para. 3.52].

The two earnings scenarios give £11.4bn and £11.9bn in 2039-40. They are averages over paths from the model that failed the test, carry the same failed condition, and are reported because the registered rule keeps them as scenarios whatever the verdict, not because they are more reliable than the average it withdrew. They are not a range around an unshifted case, and their midpoint is not an estimate of the withdrawn average. Their margins overlap, and as shares of basic and new State Pension spending they are both 4.4% (log gaps of 4.48 and 4.50 points), so the two cannot be told apart in 2039-40: most of the cash difference comes from the larger pension base that faster nominal earnings growth brings. Earlier the scenarios differ. In 2034-35 the saving is £4.4bn with higher earnings growth against £3.1bn with lower, 2.2% against 1.6% of spending: larger earnings leads use up the adjusted rule's excess over its earnings path sooner, so the increment in Section 3.3 brings the saving forward.

Table 3: Reduction in State Pension spending under the adjusted Triple Lock relative to the triple lock on four scenarios, £ billion, UK, cash terms, and in the final year as a share of basic and new State Pension spending under the triple lock, with the log gap G between the rules (log points).

	2034-35		2039-40			
Scenario	Gross	Net	Gross	Net	Share of spending (%)	G
OBR central path	0.4	0.3	0.5	0.3	0.19	0.19
OBR long-term path (triple lock at the OBR's long-term rate from April 2032)	3.1	2.1	10.9	7.3	4.25	4.34
Earnings −0.5 points from 2031, average of 160 paths	3.1 ± 0.4	2.2 ± 0.3	11.4 ± 0.5	7.8 ± 0.3	4.38 ± 0.19	4.48
Earnings +0.5 points from 2031, average of 160 paths	4.4 ± 0.5	3.0 ± 0.3	11.9 ± 0.5	8.1 ± 0.3	4.40 ± 0.18	4.50

Note: The OBR central path follows the OBR's October 2026 forecast to 2030 and its March 2026 long-term projection after. The OBR central and long-term paths are single full runs. The earnings scenarios are stratified estimates over the same 160 simulated paths, each run in full, ± the 95% Monte Carlo error. A scenario envelope, not a probability interval.

5.2 The OBR central path

The OBR central path is one smooth path. The floor sets the triple lock in April 2029, 2030 and 2031; from April 2032 earnings lead every year (Figure 4). The two rules differ in one year, April 2032: after the floor years the adjusted Triple Lock rises only by the floor, or by what brings it back to its earnings path, while the triple lock rises with earnings from the higher level. By 2039-40 the full new State Pension is £18,876 a year under the triple lock and £18,840 under the adjusted rule, and the saving is £0.5bn (£0.3bn net). The OBR finds the same for its forecast period: "in our current medium-term forecast an uprating of 2.5 per cent would apply under both the previous and adjusted version of the triple lock in both 2030-31 and 2031-32" [Office for Budget Responsibility, 2026a, para. 3.50, footnote 17].

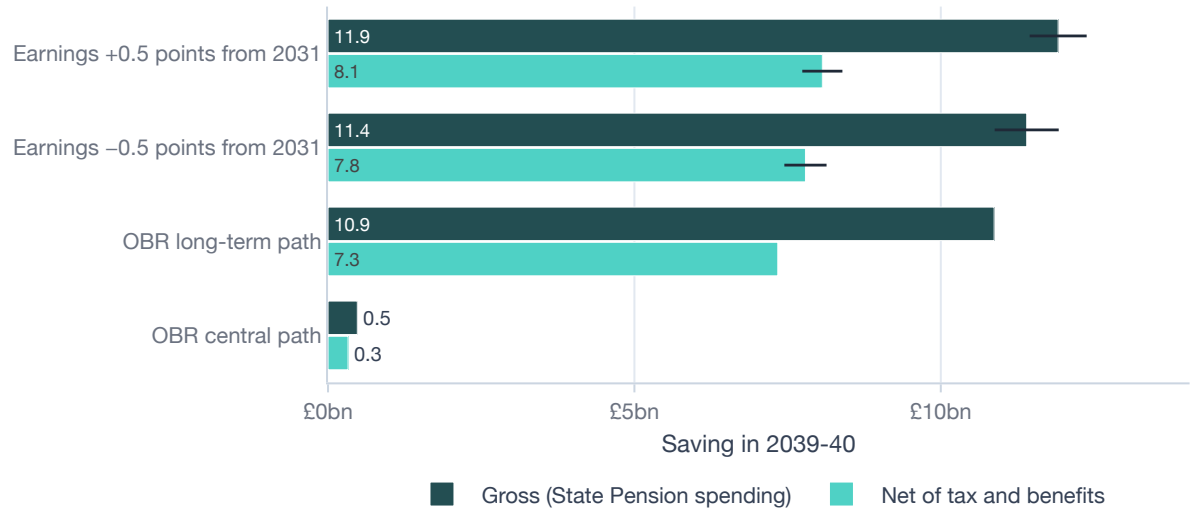


Figure 3: The saving depends on the path: small on the OBR's smooth central path, far larger where prices and earnings swap the lead or the triple lock outpaces earnings. Gross and net saving in the final year on each scenario, £ billion; whiskers are the 95% Monte Carlo error on the draw averages. The OBR central path follows the OBR's October 2026 forecast to 2030 and its March 2026 long-term projection after.

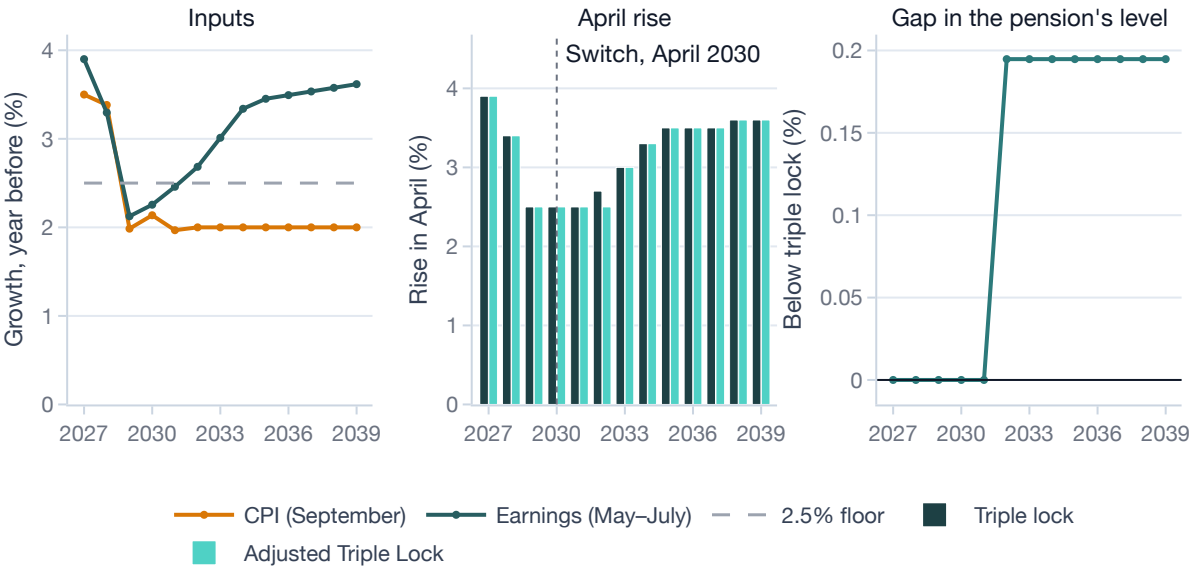


Figure 4: On the OBR central path the floor sets the early rises, and the adjusted Triple Lock differs from the triple lock in one April, 2032; the gap then persists. Left: the inputs behind each April's rise, shown at the April they set. Centre: the April rise under each rule; the dotted line marks the switch. Right: how far the adjusted rule's full new State Pension sits below the triple lock's.

5.3 The mechanism on simulated paths

We run three simulated paths in full: one drawn at random, and two picked near given percentiles of the gap between the rules in 2039-40 among the 50,000 draws, one at the 48th percentile and one at the 88th. The random path's gap, £19.57 a week, lies between the 50th and 75th percentiles. Each is one path the model makes, not a forecast, and carries no probability weight. Figure 5 shows how far apart the two rules end on each, beside the two OBR scenarios, and Table 4 gives the saving, the changes of lead and the Aprils with earnings growth below the floor on each. Most of a path's saving comes from a few years of high inflation or weak earnings followed by years in which earnings lead, after which the gap stays. The count of changes of lead does not order the savings, because the increment depends on the size of each lead as well as its timing: the path near the 90th percentile has fewer changes of lead than the path near the median, but a longer run of earnings-led years after its price-led ones. The random path's pension is higher under both rules because it draws higher CPI and earnings growth than the OBR's forecast, which also raises the spending base its saving is drawn from.

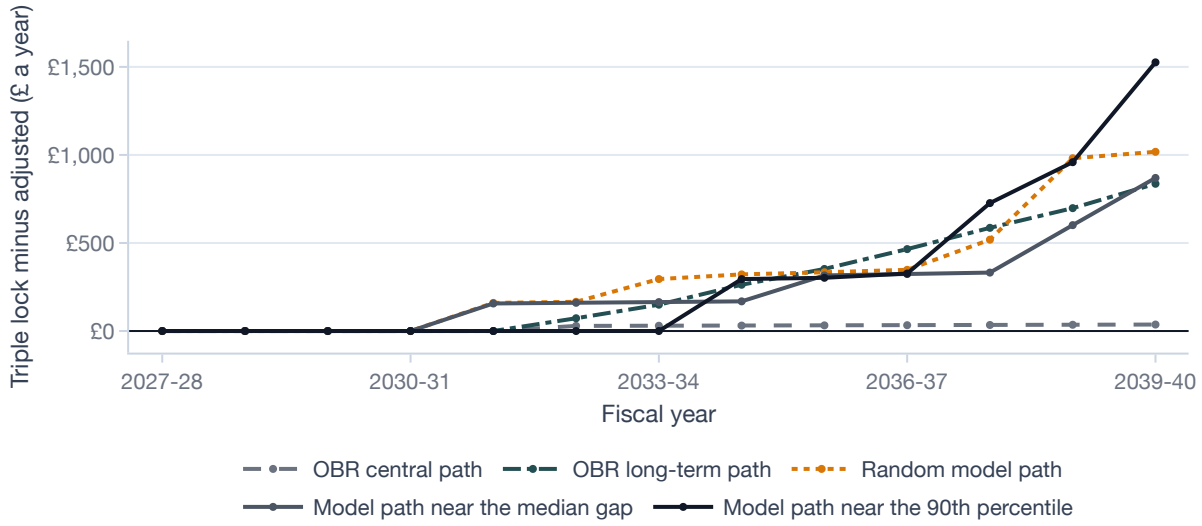


Figure 5: On simulated paths the gap opens in a few steps and then persists. How far the adjusted Triple Lock's full new State Pension ends below the triple lock's on each path run in full, £ a year.

5.4 Distributional effects

The household results run each path through PolicyEngine UK for the whole survey population, once under each rule. Every pensioner's State Pension is recalculated, and with it their income tax, Pension Credit, Housing Benefit, council tax reduction and Winter Fuel Payment. We report them on the OBR long-term path, the largest of the single runs, and give the poverty changes on the OBR central path and the two simulated paths picked by their gap beside it. Cash and percentage changes in income scale roughly with the gap in the pension; poverty rates do not.

Table 4: The paths run in full, in the final year: the full new State Pension under each rule, the saving, the average loss among households with less income than under the triple lock, and, over the Aprils from the switch to the final year, the changes of lead between earnings and the higher of CPI and the floor and the Aprils with May–July earnings growth below the floor (rule arithmetic).

Path	Triple lock (£ a year)	Adjusted (£ a year)	Gross (£bn)	Net (£bn)	Average loss (£ a year)	Changes of lead	Aprils below the floor
OBR central path	£18,876	£18,840	0.48	0.33	£35	one	two
OBR long-term path	£19,676	£18,840	10.87	7.34	£801	one	two
Random model path	£21,045	£20,028	13.23	9.06	£973	five	two
Model path near the median gap	£18,782	£17,912	11.30	7.69	£836	five	six
Model path near the 90th percentile	£19,630	£18,104	19.83	13.45	£1,459	three	five

Note: On every path about the same share of households, those receiving a State Pension, has less income.

5.5 From gross to net

Table 5 follows the gross saving on the OBR long-term path to the net saving. In 2039-40 extra Pension Credit offsets £0.81bn of the £10.87bn gross saving and lower income tax £2.39bn, leaving £7.34bn net, 68% of the gross. The net figure counts direct taxes and benefits only, with no change in behaviour. It leaves out VAT and other indirect taxes on pensioners' lower spending, which would make it smaller. Not every offset falls on the same budget: council tax reduction is paid by local authorities, and part of the income tax would otherwise go to the Scottish and Welsh governments.

Table 5: From gross to net on the OBR long-term path in the final year, £ billion.

	2039-40 (£bn)
Gross saving (State Pension)	10.87
Extra Pension Credit	−0.81
Extra Housing Benefit	−0.23
Extra council tax reduction	−0.05
Extra Winter Fuel Payment	−0.03
Less income tax	−2.39
Other taxes and benefits	−0.03
Net saving	**7.34**

Note: Each offset is signed so the column adds up to the net saving.

5.6 Who has less income

On the OBR long-term path households with less income under the adjusted Triple Lock than under the triple lock in 2039-40 have £801 a year less on average, against £35 on the OBR central path. Single pensioners have 0.8% less income than under the triple lock on average, and pensioner couples 1.3%. The share of households with at least £1 a year less income, about 30% on every path, counts the households that receive a State Pension and says nothing about the size of the effect.

Figure 6 ranks households by equivalised household net income before housing costs under the triple lock, with each decile holding a tenth of people, and gives each decile's total change

in income as a share of its total income. Relative to income, the change is largest in the 3rd decile, at 0.56% of income, and smaller in the lowest-income and in the higher-income deciles. The guarantee credit, which makes up a lower pension pound for pound, and the State Pension's smaller share of higher incomes both work in this direction.

On the OBR central path, where the pension gap is small, no poverty measure changes by more than 0.03 percentage points. On the OBR long-term path pensioner poverty after housing costs is 0.70 points higher against the fixed line and 0.75 points higher against the relative one, and on the simulated path with the largest gap up to 1.52 points higher (Table 6). Poverty among everyone changes much less, and relative poverty among everyone falls: the relative line is 60% of median income in each run, and the lower pension lowers the median, so fewer people of working age fall below the line. The fixed line is 60% of the 2024-25 median, held fixed in real terms by uprating it with each path's CPI, so it differs between paths; it also differs from the official absolute measure in Households Below Average Income statistics, which is anchored in 2010-11. Pension Credit makes up much of a lower pension for pensioners on the guarantee, but not for those just above it.

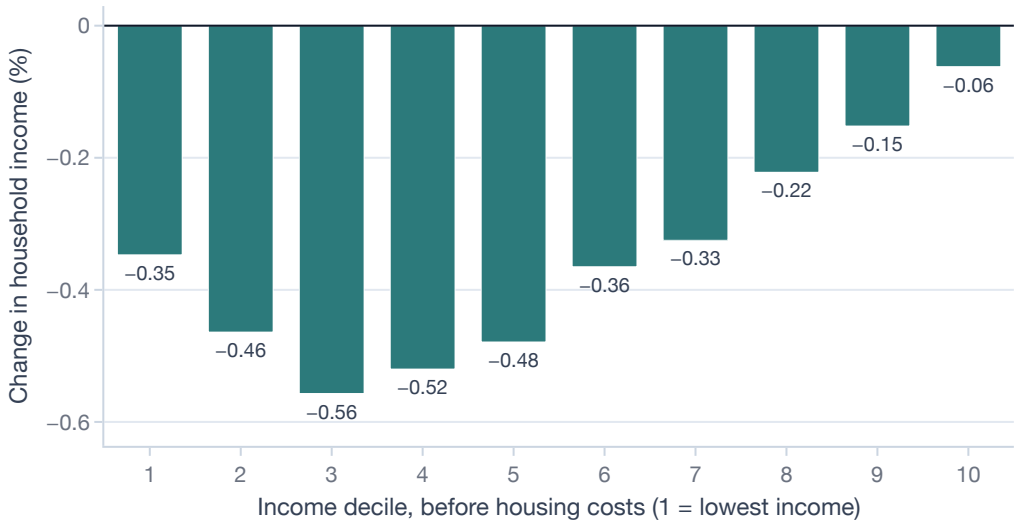


Figure 6: The loss is largest, relative to income, in the lower-middle deciles. Each decile's total change in household net income under the adjusted Triple Lock against the triple lock, as a % of its total income, on the OBR long-term path in the final year, by decile of equivalised household net income before housing costs under the triple lock (each decile holds a tenth of people).

5.7 Example pensioners

Four hypothetical pensioners, run as households under each rule on the model path near the median gap, show how tax and benefits pass on a lower State Pension (Table 7, Figure 7); the OBR long-term run carries no example households. Each receives the full flat-rate State Pension and, unlike the survey runs, claims every benefit it is entitled to. A lower State Pension means less income tax for a pensioner above the personal allowance and more Pension Credit for one entitled to it; Pension Credit counts income after tax [State Pension Credit Regulations, 2002],

Table 6: Poverty after housing costs in the final year: share of people in households below the line, %, and the change under the adjusted Triple Lock, percentage points.

Measure	Triple lock (%)	Adjusted (%)	Change, OBR central (points)	Change, OBR long term (points)	Change, median-gap path (points)	Change, 90th-percentile path (points)
Pensioners, fixed line	6.73	7.43	0.01	0.70	0.88	1.52
Pensioners, relative	10.62	11.37	0.03	0.75	0.66	0.93
Everyone, fixed line	15.89	16.06	0.00	0.17	0.20	0.36
Everyone, relative	19.28	19.12	−0.01	−0.16	−0.03	−0.23

Note: Levels are for the OBR long-term path only. Fixed line: below 60% of the 2024-25 median, uprated by each path's CPI, so the line differs between paths. Relative: below 60% of the median in the same run.

and council tax reduction disregards the income of a pensioner receiving the guarantee credit [Council Tax Reduction Schemes (Prescribed Requirements) (England) Regulations, 2012]. The last row of Table 7 gives the share of the lower pension that reaches each pensioner's income in 2039-40.

Thresholds explain the differences, and two modelling conventions set the thresholds (Section 4). The Pension Credit guarantee rises with earnings under both rules, and the personal allowance stays at £12,570 to 2030-31 and then rises with CPI, so the full new State Pension is above the allowance and taxed once the rules can differ. With only the full new State Pension, a pensioner sits close to the guarantee after tax. Under the adjusted rule the pension (a) therefore receives more guarantee credit than under the triple lock in some years (2031-32, 2032-33, 2035-36 and 2039-40), and with it more council tax reduction. In those years more of the lower pension is made up, which is why the line in Figure 7 moves from year to year. The pensioner with a £15,000 private pension (b) is above every threshold and bears most of the loss, less income tax. The council tenant (c) is better off under the adjusted rule in 2039-40 because of a cliff: the lower pension takes the tenant onto the guarantee credit, which passports maximum Housing Benefit, £1,334 a year more, and full council tax reduction, and together with less income tax these more than make up the lower pension. The survey-wide results cannot reproduce such a gain, because there Housing Benefit and council tax reduction respond only for households already receiving them (Section 4). The older pensioner on the basic State Pension (d) is on the guarantee under both rules: extra guarantee credit makes up the lower pension in full, but they receive £272 less savings credit. Whether (a) and (c) reach the guarantee depends on the conventions: if the guarantee instead rose by the same percentage as the State Pension under each rule, if the personal allowance were higher or if a successor to the Pension Tax Allowance removed their income tax, they would be less likely to cross it, and (a) and (c) would lose the income tax offset.

Table 7: Example pensioners on the model path near the median gap in the final year: the change in each item of income under the adjusted Triple Lock, £ a year (a positive figure adds to income).

	(a)	(b)	(c)	(d)
State Pension	−£869	−£869	−£869	−£666
Income tax (less paid)	£174	£174	£174	£0
Pension Credit	£235	£0	£235	£394
Housing Benefit	£0	£0	£1,334	£0
Council tax reduction	£410	£0	£410	£0
Winter Fuel Payment	£0	£0	£0	£0
Income after tax and benefits	−£50	−£696	£1,284	−£272
Share of the State Pension fall that reaches income	6%	80%	−148%†	41%

Note: The last row is the change in income after tax and benefits as a share of the change in State Pension. † A negative share: income rises although the pension falls.
The examples: (a) single pensioner aged 70, State Pension only, owns their home; (b) single pensioner aged 70 with a £15,000 private pension, owns their home; (c) single pensioner aged 70, State Pension only, rents from the council at £120 a week; (d) single pensioner aged 90 on the old basic State Pension, rents from the council at £120 a week.

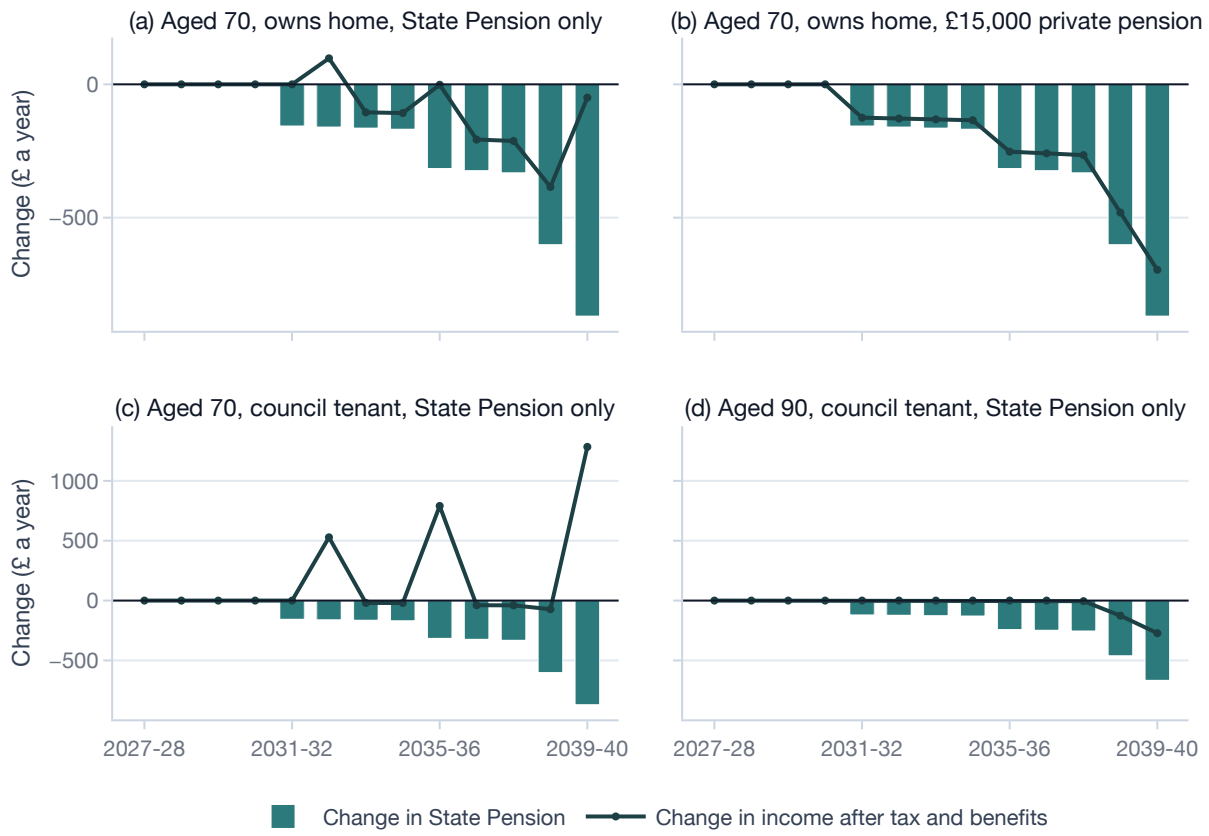


Figure 7: Benefits make up a varying share of the lower pension, depending on where each pensioner sits against the Pension Credit and other thresholds. The change in State Pension under the adjusted Triple Lock (bars) and in income after tax and benefits (line), £ a year, on the model path near the median gap.

6 Comparison with official costings

DWP costs the adjusted Triple Lock at £15bn in 2039-40 in cash terms, or £11bn in 2025-26 prices [Department for Work and Pensions, 2026b], and the Budget repeats the cash figure (Section 1). It is direct spending for Great Britain, rounded to the nearest billion, and it leaves out tax and debt interest. DWP's method note says that "Pensim3 uses administrative and survey data, including the Family Resources Survey, to project the population and individual pension outcomes from 2018 to 2100. The analysis in this publication covers Great Britain", that its estimates use the ONS's 2024-based national population projections, and that "Components of the pre-2016 State Pension that are not covered by the Triple Lock are assumed to continue to be uprated by CPI" [Department for Work and Pensions, 2026b]. DWP's model is a dynamic microsimulation model of the kind Li and O'Donoghue [2013] review. The Budget states that "The government's central forecast for the trajectory of the State Pension and how it differs from the Triple Lock is set out below" [HM Treasury, 2026, Box 5.A], but it publishes neither the inputs of that trajectory nor its yearly values. The government has also said, in answer to a parliamentary question, that DWP's analysis did not use single-year earnings or prices growth assumptions [UK Parliament, 2026], so DWP's figure does not correspond to any one of our paths.

Our nearest figures are the gross savings in Great Britain: £0.5bn on the OBR central path, £10.6bn on the OBR long-term path, and £11.1bn and £11.6bn on the two earnings scenarios. Deflated by the OBR central path's CPI, the last two are £8.2bn and £8.6bn in 2025-26 prices, beside DWP's £11bn. The population also differs. DWP projects deaths, migration and State Pension entitlements, while we reweight today's survey (Section 4.2). Our survey's spending on the basic and new State Pension, the base the saving scales with, is 94% of DWP's forecast for 2026-27 (Table 8); DWP's figure includes pensions paid abroad that rise under either rule, while State Pensions paid in countries without uprating agreements do not rise and are unaffected by either rule. Since the gross saving is close to $(1 - e^{-G_t})B_t^{\mathrm{TL}}$, scaling our base up to DWP's in that year would raise the Great Britain figures in proportion, to £11.3bn on the OBR long-term path and £11.8bn and £12.3bn on the earnings scenarios. This separates the uprating rule from the population base, but not the effect of later cohorts' larger new State Pension entitlements, which our base also understates. DWP itself cautions against reading single-year point estimates from its figures, and its saving keeps growing, to £50bn in 2049-50; our horizon ends in 2039-40.

The OBR's own figures measure related but different things. Its footnote on the medium-term forecast matches our OBR central path, on which the adjusted rule rises by 2.5% in April 2030 and 2031 [Office for Budget Responsibility, 2026a, para. 3.50, footnote 17]. Its replay of 1990–2025 puts payments under the adjusted rule 1.4% lower on average than under the triple lock, an average over all years with the anchor set in 1990; the same average on our replay, which starts in April 2012, is 1.6% (Section 3.4). Its estimate that the triple lock will have added £15.5 billion a year by 2029-30 compares the triple lock with earnings uprating [Office for Budget Responsibility, 2025], a different counterfactual from the adjusted rule, which keeps the protection against prices and the floor; our nearest figure, the adjusted rule from April 2012, saves £7.8bn in 2026-27.

DWP also projects pensioner relative poverty after housing costs under the adjusted Triple Lock falling from around 14% to around 8% by 2049-50, against a poverty line projected forward with earnings. Our poverty rates (Section 5.4) are for 2039-40, use a line set each year from the simulated incomes, and compare the adjusted Triple Lock with the triple lock rather than with today, so they answer a different question.

7 Discussion

Three points bear on how max-of-indices rules are costed. First, on a path on which earnings lead in almost every year, as on the OBR's central path, the decomposition of Section 3.3 gives a gap near zero, while paths whose inputs vary from year to year give a much larger saving, with average earnings growth 0.5 points either side of the OBR's. This does not show that any official long-term costing understates the saving: the OBR notes that the change does not affect its forecast period and uses a premium for the long term, DWP uses its own projection, and both say that the result depends on the order and covariation of the indices (Section 1). What the paper adds is the size of that dependence on stochastic paths with full microsimulation, and the exact decomposition behind it. The comparison between the central path and the earnings scenarios also mixes two things, since the scenarios shift mean earnings growth as well as varying it from year to year.

Second, the OBR's long-term premium is a reduced-form allowance for the same effect. Applying it every year adds a lead for prices or the floor in every year, which no path of the inputs produces, and gives the adjusted rule no premium of its own, so the OBR long-term path is an upper bound under that costing convention, not a forecast.

Third, the test does not sign the bias in the simulated savings. The main version expected earnings growth below the floor in 44.6% of years against 17.9% in practice, and such years set $d_t > 0$. But a floor year adds to the gap only when earnings lead later, and the increment in Section 3.3 is larger when earnings leads are larger, so fewer floor years with stronger earnings can open the gap faster; the earnings scenarios show the saving rising with earnings growth through the mid-2030s. The main version's lead changes were close to those realised on the test that decides the verdict (1.06 against 1.00 per four-year window), and on the paths run in full the number of changes of lead does not order the savings (Table 4). The sign of the bias in the simulated savings is therefore not identified, and a model that passed the test could give higher or lower averages.

8 Conclusion

The adjusted Triple Lock keeps the triple lock's protection against prices and its 2.5% floor, but lets the pension fall back to its April 2029 value relative to earnings after years in which prices or the floor beat earnings. Its saving therefore depends on how often the lead passes from prices or the floor to earnings, not only on their expected growth: it grows only in earnings-led years that follow a price- or floor-led one. Replayed from April 2012 on today's figures, it would leave the pension 6.2% below the triple lock in 2026-27. On the OBR's smooth central path it lowers State Pension spending by £0.5bn in 2039-40; under the OBR's long-term triple-lock convention, an upper bound under that convention, by £10.9bn; and averaged over simulated paths with earnings growth 0.5 points lower or higher by £11.4bn or £11.9bn, £7.8bn or £8.1bn after tax and benefits respond. These are scenarios, not a range with probabilities attached: our model of prices and earnings failed a registered test on past forecasts by overstating how often earnings growth falls below the floor, so the paper gives no expected value. DWP's £15bn for Great Britain rests on a path it does not publish and is not comparable like for like.

The loss falls on households that receive a State Pension, with Pension Credit and income tax making up part of it. On the OBR long-term path pensioner poverty after housing costs rises by 0.70 percentage points against the fixed line and 0.75 against the relative one. The rule

would also require a change to the statutory earnings link. For any uprating rule that takes the highest of several indices, the cost of removing the ratchet depends on how the indices vary from year to year, and a test of the model of that variation decides what can be reported: here, scenarios. A test on outcomes not yet observed can first say, in October 2034, whether averages over the paths may be reported as estimates.

The analysis has limits a reader should weigh. It gives no expected value, the sign of the failed condition's effect on the scenario averages is not identified, and the simulated paths rest on one model of monthly prices and earnings, centred on the OBR's forecasts, with no periods of changing volatility. The population is reweighted to projected totals rather than aged, its basic and new State Pension spending in Great Britain is 94% of DWP's forecast, and projected spending in 2039-40 is not checked against the OBR's long-term projection. Behaviour is held fixed, Housing Benefit and council tax reduction respond only for households already receiving them, and indirect taxes are left out, so the net figures count direct taxes and benefits only.

References

Vahid Ahmadi, Max Ghenis, Nikhil Woodruff, Pavel Makarchuk, and Anthony Volk. policyengine: a microsimulation tool for tax-benefit policy analysis. *Journal of Open Source Software*, 11(125):11115, 2026. doi: 10.21105/joss.11115.

Erik Britton, Paul Fisher, and John Whitley. The inflation report projections: understanding the fan chart. *Bank of England Quarterly Bulletin*, 38(1):30–37, 1998. URL https://www.bankofengland.co.uk/quarterly-bulletin/1998/q1/the-inflation-report-projections-understanding-the-fan-chart.

Katherine Casey, Rachel Glennerster, and Edward Miguel. Reshaping institutions: evidence on aid impacts using a preanalysis plan. *Quarterly Journal of Economics*, 127(4):1755–1812, 2012. doi: 10.1093/qje/qje027.

Oya Celasun, Xavier Debrun, and Jonathan D. Ostry. Primary surplus behavior and risks to fiscal sustainability in emerging market countries: a "fan-chart" approach. *IMF Staff Papers*, 53(3):401–425, 2006. doi: 10.2307/30035919.

Garret Christensen and Edward Miguel. Transparency, reproducibility, and the credibility of economics research. *Journal of Economic Literature*, 56(3):920–980, 2018. doi: 10.1257/jel.20171350.

Todd E. Clark. Real-time density forecasts from Bayesian vector autoregressions with stochastic volatility. *Journal of Business & Economic Statistics*, 29(3):327–341, 2011. doi: 10.1198/jbes.2010.09248.

Council Tax Reduction Schemes (Prescribed Requirements) (England) Regulations. SI 2012/2885, Schedule 1, Part 6, Chapter 2, paragraph 13, 2012. URL https://www.legislation.gov.uk/uksi/2012/2885/schedule/1/paragraph/13. Accessed 16 October 2026.

Frank de Jong. Pension fund investments and the valuation of liabilities under conditional indexation. *Insurance: Mathematics and Economics*, 42(1):1–13, 2008. doi: 10.1016/j.insmatheco.2006.11.004.

Department for Work and Pensions. Benefit expenditure and caseload tables 2026. Outturn and forecast tables: Spring Forecast 2026, April 2026a. URL https://www.gov.uk/government/publications/benefit-expenditure-and-caseload-tables-2026. Published 14 April 2026. Accessed 16 October 2026.

Department for Work and Pensions. State Pension uprating. Research and analysis, in collection "state pension uprating analysis 2026", Department for Work and Pensions, September 2026b. URL https://www.gov.uk/government/publications/state-pension-uprating-analysis-2026/state-pension-uprating. Published 29 September 2026. Accessed 16 October 2026.

Francis X. Diebold, Todd A. Gunther, and Anthony S. Tay. Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4):863–883, 1998. doi: 10.2307/2527342.

John Geanakoplos and Stephen P. Zeldes. Market valuation of accrued Social Security benefits. In Deborah Lucas, editor, *Measuring and Managing Federal Financial Risk*, pages 213–234. University of Chicago Press, 2010. doi: 10.7208/chicago/9780226496597.003.0009.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(2):243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.

M. Barry Goldman, Howard B. Sosin, and Mary Ann Gatto. Path dependent options: "buy at the low, sell at the high". *Journal of Finance*, 34(5):1111–1127, 1979. doi: 10.1111/j.1540-6261.1979.tb00059.x.

HM Treasury. Budget 2026. Technical report, HM Treasury, October 2026. URL https://www.gov.uk/government/publications/budget-2026-document/budget-2026-html. Published 28 October 2026; paragraph 5.23 and Box 5.A (the rule), paragraph 1.29 (the saving). Accessed 5 November 2026.

Markus Knell. How automatic adjustment factors affect the internal rate of return of PAYG pension systems. *Journal of Pension Economics and Finance*, 9(1):1–23, 2010. doi: 10.1017/S1474747209990035.

Ronald Lee and Shripad Tuljapurkar. Uncertain demographic futures and social security finances. *American Economic Review*, 88(2):237–241, 1998. URL https://ideas.repec.org/a/aea/aecrev/v88y1998i2p237-41.html. Papers and Proceedings.

Jinjing Li and Cathal O'Donoghue. A survey of dynamic microsimulation models: uses, model structure and methodology. *International Journal of Microsimulation*, 6(2):3–55, 2013. doi: 10.34196/ijm.00082.

D. V. Lindley. The theory of queues with a single server. *Mathematical Proceedings of the Cambridge Philosophical Society*, 48(2):277–289, 1952. doi: 10.1017/S0305004100027638.

Deborah Lucas. Valuation of government policies and projects. *Annual Review of Financial Economics*, 4:39–58, 2012. doi: 10.1146/annurev-financial-110311-101829.

Robert C. Merton. An analytic derivation of the cost of deposit insurance and loan guarantees: an application of modern option pricing theory. *Journal of Banking & Finance*, 1(1):3–11, 1977. doi: 10.1016/0378-4266(77)90015-2.

James Mitchell and Kenneth F. Wallis. Evaluating density forecasts: forecast combinations, model mixtures, calibration and sharpness. *Journal of Applied Econometrics*, 26(6):1023–1040, 2011. doi: 10.1002/jae.1192.

OECD. *Pensions at a Glance 2023: OECD and G20 Indicators*. OECD Publishing, Paris, 2023. doi: 10.1787/678055dd-en.

Office for Budget Responsibility. Fiscal risks and sustainability – July 2025. Technical report, Office for Budget Responsibility, July 2025. URL https://obr.uk/frs/fiscal-risks-and-sustainability-july-2025/. Published 8 July 2025. Accessed 16 October 2026.

Office for Budget Responsibility. Economic and fiscal outlook – October 2026. Technical report, Office for Budget Responsibility, October 2026a. URL https://obr.uk/efo/economic-and-fiscal-outlook-october-2026/. Published 28 October 2026; paragraphs 3.48–3.53. Accessed 5 November 2026.

Office for Budget Responsibility. Historical official forecasts database – March 2026. Spreadsheet (Historical_official_forecasts_database_Spring_2026.xlsx), March 2026b. URL https://obr.uk/docs/dlm_uploads/Historical_official_forecasts_database_Spring_2026.xlsx. Listed at https://obr.uk/data/, dated 26 March 2026. Accessed 16 October 2026.

Office for Budget Responsibility. Long-term economic determinants – March 2026 Economic and fiscal outlook. Spreadsheet, dated 28 May 2026 (sheets "Long-term economic determinants" and "Growth rate assumptions"), May 2026c. URL https://obr.uk/docs/dlm_uploads/Long-term-economic-determinants-March-2026-EFO.xlsx. Accessed 16 October 2026.

Office for National Statistics. CPI INDEX 00: ALL ITEMS 2015=100. Time series D7BT, dataset MM23, release of 21 October 2026, October 2026a. URL https://www.ons.gov.uk/economy/inflationandpriceindices/timeseries/d7bt/mm23. Accessed 5 November 2026.

Office for National Statistics. CPI ANNUAL RATE 00: ALL ITEMS 2015=100. Time series D7G7, dataset MM23, release of 21 October 2026, October 2026b. URL https://www.ons.gov.uk/economy/inflationandpriceindices/timeseries/d7g7/mm23. Accessed 5 November 2026.

Office for National Statistics. AWE: Whole Economy Level (£): Seasonally Adjusted Total Pay Excluding Arrears. Time series KAB9, dataset LMS, release of 20 October 2026, October 2026c. URL https://www.ons.gov.uk/employmentandlabourmarket/peopleinwork/earningsandworkinghours/timeseries/kab9/lms. Accessed 5 November 2026.

Office for National Statistics. AWE: Whole Economy Year on Year Three Month Average Growth (%): Seasonally Adjusted Total Pay Excluding Arrears. Time series KAC3, dataset LMS, release of 20 October 2026, October 2026d. URL https://www.ons.gov.uk/employmentandlabourmarket/peopleinwork/earningsandworkinghours/timeseries/kac3/lms. Accessed 5 November 2026.

Office for National Statistics. National population projections: 2024-based, UK, principal projection, 2026e. URL https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationprojections/datasets/z1zippedpopulationprojectionsdatafilesuk. Released 28 April 2026. Accessed 16 October 2026.

Benjamin A. Olken. Promises and perils of pre-analysis plans. *Journal of Economic Perspectives*, 29(3):61–80, 2015. doi: 10.1257/jep.29.3.61.

Pensions Act. 2014 c. 19, Part 1, section 1(2): a person who reaches pensionable age before 6 April 2016 is not entitled to the state pension under that Part, 2014. URL https://www.legislation.gov.uk/ukpga/2014/19/section/1. Accessed 16 October 2026.

PolicyEngine. Microcosm UK 2024-25, national cut. Dataset, national cut released with policyengine 6.11.1, October 2026a. URL https://huggingface.co/datasets/policyengine/populace-uk-private. Access-controlled.

PolicyEngine. policyengine. Python package, version 6.11.1, October 2026b. URL https://pypi.org/project/policyengine/6.11.1/. Source: https://github.com/PolicyEngine/policyengine.py. Accessed 16 October 2026.

PolicyEngine. policyengine-uk. Python package, version 2.151.0, October 2026c. URL https://pypi.org/project/policyengine-uk/2.151.0/. Source: https://github.com/PolicyEngine/policyengine-uk. Accessed 16 October 2026.

Jonathan Portes. Unlocking the pensions debate: the origins and future of the 'triple lock'. *Political Quarterly*, 94(4):668–673, 2023. doi: 10.1111/1467-923X.13330.

Stephen Pudney, Ruth Hancock, and Holly Sutherland. Simulating the reform of means-tested benefits with endogenous take-up and claim costs. *Oxford Bulletin of Economics and Statistics*, 68(2):135–166, 2006. doi: 10.1111/j.1468-0084.2006.00156.x.

Matteo Richiardi, Diego Collado, and Daria Popova. UKMOD – a new tax-benefit model for the four nations of the UK. *International Journal of Microsimulation*, 14(1):92–101, 2021. doi: 10.34196/ijm.00231.

Michael Scheuerer and Thomas M. Hamill. Variogram-based proper scoring rules for probabilistic forecasts of multivariate quantities. *Monthly Weather Review*, 143(4):1321–1334, 2015. doi: 10.1175/MWR-D-14-00269.1.

Social Security Administration Act. 1992 c. 5, section 150A, 1992. URL https://www.legislation.gov.uk/ukpga/1992/5/section/150A. Accessed 16 October 2026.

Social Security Benefits Up-rating Order. SI 2011/821, article 5(3): the basic pension in a Category A retirement pension rises from £97.65 to £102.15 a week, 2011. URL https://www.legislation.gov.uk/uksi/2011/821/made. Accessed 16 October 2026.

Social Security Benefits Up-rating Order. SI 2023/316: the full new State Pension rises from £185.15 to £203.85 a week (State Pension Regulations 2015, regulation 1A) and the single standard minimum guarantee from £182.60 to £201.05 (State Pension Credit Regulations 2002, regulation 6(1)(b)), 2023. URL https://www.legislation.gov.uk/uksi/2023/316/made. Accessed 5 November 2026.

Social Security (Up-rating of Benefits) Act. 2021 c. 32, section 1: for the tax year ending with 5 April 2022 the section 150A review is "by reference to the general level of prices" instead of earnings, 2021. URL https://www.legislation.gov.uk/ukpga/2021/32/section/1/enacted. Accessed 16 October 2026.

State Pension Credit Regulations. SI 2002/1792, regulation 17(10): "there shall be disregarded– (a) any amount payable by way of tax", 2002. URL https://www.legislation.gov.uk/uksi/2002/1792/regulation/17. Accessed 16 October 2026.

Holly Sutherland and Francesco Figari. EUROMOD: the European Union tax-benefit microsimulation model. *International Journal of Microsimulation*, 6(1):4–26, 2013. doi: 10.34196/ijm.00075.

UK Parliament. Written question HL3774. House of Lords written question, tabled 5 October 2026, answered 15 October 2026, October 2026. URL https://questions-statements.parliament.uk/written-questions/detail/2026-10-05/hl3774. Accessed 5 November 2026.

Carlos Vidal-Meliá, María del Carmen Boado-Penas, and Ole Settergren. Automatic balance mechanisms in pay-as-you-go pension systems. *Geneva Papers on Risk and Insurance – Issues and Practice*, 34(2):287–317, 2009. doi: 10.1057/gpp.2009.2.

A Proof of the proposition

From the level equations, $y_t - y_{t-1} = \ln(1 + \max(m_t, e_t)) - \ln(1 + e_t) = \max(d_t, 0)$, and $x_t = \ln \max\big((L_{t-1}/A_{t-1})(1 + m_t)/(1 + e_t),\, 1\big) = \max(x_{t-1} + d_t, 0)$. Hence $\Delta G_t = \max(d_t, 0) - \max(d_t, -x_{t-1})$, which is zero when $d_t \geq 0$ and $\min(-d_t, x_{t-1})$ when $d_t < 0$; since $x_{t-1} \geq 0$ this is $\min(\max(-d_t, 0), x_{t-1}) \geq 0$. Summing, the reflected walk has the standard solution $x_t = D_t - \min_{T_0 \leq s \leq t} D_s$, and $y_t = \sum_s \max(d_s, 0)$, which gives the closed form for G_t. The upper bounds follow from $x_t \geq 0$ and from $\min_s D_s \leq D_{T_0} = 0$.

If $\Delta G_r > 0$ then $d_r < 0$ and $x_{r-1} > 0$, and $x_{r-1} > 0$ requires some $s < r$ with $d_s > 0$, since x starts at zero and rises only when $d > 0$. Conversely, take s with $d_s > 0$ and the first $r > s$ with $d_r < 0$; between them every $d \geq 0$, so $x_{r-1} \geq x_s \geq d_s > 0$ and $\Delta G_r > 0$. So $G_t > 0$ exactly when such a pair $s < r \leq t$ exists. If $d_s \leq 0$ in every year, x stays at zero and $G_t = y_t = 0$; if $d_s \geq 0$ in every year, no increment is positive and $G_t = 0$.

B Further results

This appendix gives the data against DWP's benefit expenditure and caseload forecasts [Department for Work and Pensions, 2026a] (Table 8) and the scores of each version of the model of prices and earnings in the test on past forecasts (Table 9 and Table 10).

Table 8: Our data against DWP's Spring Forecast 2026, Great Britain.

		Microcosm		Enhanced FRS 1.58.2	
2026-27	DWP	Value	% of DWP	Value	% of DWP
State Pension, excluding payments abroad (£bn)	148.3	144.8	98%	124.8	84%
Basic and new State Pension (a) (£bn)	131.0	123.1	94%	110.6	84%
State Pension recipients (m)	12.15	12.06	99%	11.07	91%
Pension Credit (£bn)	6.1	6.4	105%	7.0	116%
Pension Credit claims (m)	1.31	1.39	106%	1.46	112%
Housing Benefit, all ages (b) (£bn)	12.5	9.3	74%	8.2	66%
Housing Benefit, pension age (£bn)	7.3	7.2	99%	7.6	104%
Housing Benefit claims, pension age (m)	1.08	1.09	100%	1.11	103%

Note: Microcosm is the dataset every result uses; the Enhanced FRS is a check only. Claims are counted in benefit units. (a) DWP's figure includes payments abroad, which ours do not. (b) Excludes Universal Credit housing costs.

Table 9: The test on past forecasts, on the Budget's inputs, with April 2022 scored as in law: coverage, accuracy scores and verdict.

	Seven forecasts		Thirteen forecasts				
Version	Yearly	Four-year	Yearly	Four-year	Plausible range	Score ratio	Verdict
Monthly (main version)	83%	5 of 7	90%	11 of 13	36% to 92%	1.00	Fail
Monthly, two lags	67%	5 of 7	73%	11 of 13	36% to 92%	1.07	Pass
Annual	79%	5 of 7	83%	11 of 13	36% to 92%	1.15	Fail
Monthly, heavy tails	63%	5 of 7	71%	11 of 13	36% to 92%	1.07	Pass
Monthly, normal shocks	83%	5 of 7	90%	11 of 13	36% to 92%	1.03	Fail

Note: Seven forecasts: one a year, March 2016 to March 2022; thirteen forecasts: June 2010 to March 2022. Yearly: share of yearly gaps inside each version's central 80% band; four-year: four-year gaps inside it; plausible range: the 95% range of four-year coverage on the seven-forecast test, which must include 80%. Score ratio: the largest ratio of the version's accuracy scores to the main version's. The rule lets only the main version supply an average.

Table 10: The test on past forecasts, continued: the bias conditions on the seven-forecast test, with April 2022 scored as in law, and the past-years check.

Version	Floor years (%)	Floor bias	Lead changes	Lead bias	Gap bias	Past years
Monthly (main version)	44.6%	0.267	1.06	0.06	1.00	55th
Monthly, two lags	42.1%	0.243	1.03	0.03	1.00	69th
Annual	45.0%	0.271	0.85	−0.15	0.98	51st
Monthly, heavy tails	42.6%	0.247	1.02	0.02	1.00	72nd
Monthly, normal shocks	45.4%	0.275	1.09	0.09	1.00	54th

Note: Floor years: the share of years with earnings growth below the floor that the version expected (realised: 17.9%); floor bias is the difference (limit 0.25). Lead changes: expected per four-year window (realised: 1.00); lead bias is the difference (limit 1.0). Gap bias: average error in earnings minus CPI, points (limit 1.5). Past years: the percentile of the realised gap since 2011 among the version's paths (limits in Table 2).